

Exploring Human-AI Interactions Across Various Learning Tasks: Diverse Perspectives on Augmenting Learner

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Abstract: This symposium focuses on understanding learners' interactions with artificial intelligence (AI), particularly chatbots and agents, to advance human-AI collaboration research from the perspective of computer-supported collaborative learning (CSCL). The four papers discuss human-AI interactions in various learning tasks and domains, especially from an augmentation perspective. The studies examine how these interactions can support learning processes and how learners perceive AI as an interaction partner (or not). By exploring these augmentation dynamics across tasks and domains, the symposium contributes to the evolving understanding of human-AI collaboration in teaching and learning. The symposium's discussant will synthesize insights from these studies to shed light on how these studies contribute knowledge and how these approaches need to be further developed to contribute to future hybrid intelligence (HI) systems in education.

Introduction

The advancements in artificial intelligence (AI) and its increased integration into educational contexts offer both novel challenges and opportunities (Gašević et al., 2023). This evolution calls for a thorough examination of the interactive interplay between human learning and AI capabilities, particularly within the hybrid intelligence (HI) research, which explores how we can combine human intelligence with AI to create systems where humans and AI complement each other (Akata et al., 2020; Järvelä et al., 2023). This human-AI collaboration is essential for maximizing the potential benefits of AI in enhancing and augmenting human learning. We still don't know much about these opportunities in teaching and learning, mainly because the empirical research is still scarce.

It is known that AI can significantly augment human learner competencies by providing personalized learning experiences, enhancing engagement, and offering real-time feedback. For example, intelligent tutoring systems (ITS; Aleven et al., 2017) and teacher dashboards (Knoop-van Kampen & Molenaar, 2020) have taken on some teaching tasks by providing real-time insights, tailored feedback, and scaffolding (Dever et al., 2024) and adjusting task difficulty (Behera et al., 2020). Also, teacher-AI collaboration can augment human teachers by allowing for new insights into learning and teaching, which were previously not yet available to teachers (Cukurova, 2024). AI agents are also able to act as facilitators of and scaffolding for collaborative learning (Naik et al., 2024). From the learner side, successful solutions have been developed to use rule-based AI for detecting individual student learning self-regulated learning strategies and providing scaffolds according to their individual needs (Fan et al., 2022).

Less is known about human learners and learning processes in and for human-AI interaction, which focuses on understanding and improving the ways humans and AI systems can work together effectively to achieve shared goals. The studies presented in this symposium are early explorations into understanding learners'

interactions with tools such as chatbots and agents. They gradually contribute to the field of human-AI collaboration from the perspective of computer-supported collaborative learning (CSCL) research. *This symposium discusses human-AI interactions in various learning tasks and domains, especially from an augmentation perspective.* Analyses on these interactions are guided by collaborative learning theory and the augmentation perspective is considered from a) learners' interactions with AI for supporting learning process b) learners' interpretations and perceptions about AI as an interaction partner.

Understanding human learners in and for AI interaction

Technological advancements in the past ten years have expanded the opportunities for using digital data for understanding complex human learning processes (Azevedo & Gašević, 2019). These advancements have also spawned emerging research directions to examine the potential of these data for informing human-AI interactions and the design of AI systems that are sensitive to individual differences in learning processes and can adapt to and support learners as they manage their learning demands both individually and collaboratively (Molenaar, 2022). It is still always a human learner who is interacting with AI agents or chatbots. Learners' interaction with AI, their beliefs, perceptions, and interpretations about AI as an interaction partner can substantially impact their learning strategies and behaviors. For instance, learners' awareness of AI as their co-author and the original source of a text play a key role in AI-assisted academic writing (Cress & Kimmerle, 2023). Their perceptions of the AI's characteristics, such as the sources it relies on or potential limitations, can help them better understand the reasoning behind AI-generated content and set effective revision goals.

More research is needed to understand learners' interaction patterns and their qualities when working with AI, such as productivity and reciprocity within dialogues with AI agents, as well as how the patterns and sequences of these interactions influence learners' content-level understanding across various domains. Multi-modal data and advanced analytics have been instrumental in identifying patterns and frequencies associated with learning (Saint et al., 2022). While these studies have offered inductive insights into systematic trends and modeled interactions as patterns of actions, they have largely overlooked the context and purpose of these interactions, as well as learners' responses to them (Winne, 2018). To address the why and how questions, it is crucial to explore learners' perceptions and contextual understandings of their interactions with AI. These perspectives provide complementary and essential insights into how AI can enhance human learning. Accordingly, the studies in this symposium contribute by adding a qualitative understanding of human-AI interactions.

The symposium will consist of a 5-minute introduction which will address the overarching questions that the symposium seeks to answer, followed by individual presentations (15-min each) and the commentary by the discussant Mutlu Cukurova (10-min). This will allow for a 15-minute general discussion between the symposium participants and the session attendees. At the beginning of their presentation, each presenter will define how their current research advances human-AI collaboration research from the perspective of computer-supported collaborative learning (CSCL) and contributes to the development of hybrid intelligence systems in education.

Contribution 1#: Learners' perceptions of AI and peer 'co-authorship' in text revision: An exploratory study on human-AI collaboration in academic writing

Anna Radtke, Jennifer Meyer, Nikol Rummel

Introduction

In AI-assisted writing, learners have to revise texts initially drafted by a text-generative AI. This situation is similar to that of revising peer-written text passages in collaborative writing. As known from empirical research, revision of peer-written texts can not only improve the quality of the content, but also prompt learners to gain insight into new facts, arguments, or writing strategies and, thereby, develop their own competencies (Rijlaarsdam et al., 2004). In AI-assisted writing, text revision is also crucial, as AI-generated content often contains several errors and inaccuracies, such as so-called hallucinations. However, there is very little research on the underlying psychological processes in human-AI collaborative writing and, more specifically, learners' approaches when revising AI-generated content. While revising peer-written texts, learners' awareness of their peers as co-authors plays a crucial role (Bodemer & Dehler, 2011; Liu et al., 2018). Such awareness can, for instance, greatly influence what learners pay attention to during revision and the extent to which they revise. In AI-assisted writing, learners also require some kind of awareness of the AI as a text originator or even as their 'co-author' (Cress & Kimmerle, 2023). Learners' perceptions of the characteristics of an AI, such as the sources it relies on, the used algorithms and potential limitations, can help them better understand the reasoning behind AI-generated content and set effective revision goals. This study aimed to investigate learners' perceptions of AI compared to a peer co-author when revising academic texts. We determined (a) whether learners reference their 'co-author' (peer vs.

AI) during revision as an indicator of their awareness of the text originator, and (b) the content of these references regarding intentionality, perceived competencies, and collaborative strategies.

Ongoing work

In an explorative study with a control-group design, participants ($N = 8$; $f = 6$; $m = 2$, $M_{\text{(age)}} = 25.88$; $SD_{\text{(age)}} = 3.91$; all students) were instructed to imagine themselves to be in the process of revising an academic text for submission in a university course. Depending on the experimental condition (each $n = 4$), the text was labeled as peer-written or AI-generated. The actual content of the text was identical and addressed the topic ‘Relationship between sleep and life quality.’ Participants were tasked to highlight the areas of the text that they considered requiring improvement and would revise in the next step. While working on the text, the participants were asked to think aloud about anything that came to mind, without further instructions on the topics to be covered. We recorded audio and video data of the revision and transcribed according to the extended content-semantic transcription rules (Dresing & Pehl, 2018). First, we analyzed how often learners referenced their ‘co-author’ while revising the text (e.g., Alex, person, ChatGPT, AI), and second, we examined the specific content of these references, searching for keywords indicating ‘co-author’s’ intentions (e.g., want, mean), perceived competencies (e.g., well done, faulty), and collaborative strategies (e.g., ask, discuss). In the peer-condition, we found that each participant referenced their co-author multiple times while revising the text with the number of times varying considerably between two and 51 times. Participants addressed their peer as: *Alex M.* (peer’s name introduced to them in the experimental scenario), *a person*, *he/him/his*, and even *you*, suggesting that they were aware of the peer co-author during the revision. When referencing their peer co-author, participants referred to intentions (‘Alex, you just wanted to fill three pages here’), competencies (‘someone in the eleventh grade’), and collaborative regulation strategies (‘Otherwise I’ll have to ask Alex again’). In the AI-condition, we found a different pattern. Two out of four participants did not reference AI as a text originator at all. Participants did not use language that could identify their perception of the AI’s mental states or intentions, but they referenced perceived competencies of AI (i.e., by venting frustration about errors in the text).

Contribution and future direction

Our findings show that when revising a text labeled as AI-generated, learners tended to not reference AI as a ‘co-author’ to the same degree as they did in revising a text labeled as peer-written. This pattern suggests that different psychological processes might be at play and it should not be taken for granted that learners perceive AI-assisted writing as a collaborative writing process, even though it allows for a high degree of interactivity between a learner and an AI. Our findings highlight that further research is needed to determine the conceptual differences and similarities in psychological processing during human-AI collaboration in academic writing.

Contribution 2#: Help-Seeking with an AI chatbot in introductory computer science: Associations with goal orientation

Stephanie Yang, Bertrand Schneider

Introduction

Learning to program is challenging, and students in CS1 courses often seek help from various sources when stuck. In academic settings, students’ help-seeking approaches vary significantly based on their goal orientation—the underlying motivation for academic tasks (Cheong et al., 2004). Students with a mastery orientation view learning as an opportunity for growth and seek instrumental help to deepen understanding (Karabenick, 2003; Newman, 1998). In contrast, students who prioritize performance outcomes (performance-oriented) or avoiding work (work-avoidant) tend to seek executive, answer-fishing help (Pintrich, 1989).

The accessibility of conversational large language models (LLMs) as learning tools sparks new questions about student help-seeking. LLMs are now widely available to students and can provide both high-level explanations and detailed code solutions—blurring the line between seeking help to learn and seeking help to complete coursework with minimal effort. Despite growing concerns from CS1 instructors (Lau & Guo, 2023), we know little about how students use LLMs in CS courses and their underlying motivations (Yang et al., 2024).

In this study, we investigate how students interact with a custom LLM-based chatbot in an introductory CS course. We combine survey and chat log data to address two research questions: (1) How do students’ motivational constructs relate to the types of questions they ask the chatbot? and (2) How do students’ perceptions of appropriate chatbot use correspond to their actual usage behaviors? Together, these questions aim to shed light on how motivational constructs and evolving norms around LLMs in CS influence student help-seeking.

Ongoing work

We collected chat logs from students using a chatbot in a large introductory CS course with over 250 students. Beginning with the third problem set, students could consult a custom chatbot for help, but not other GenAI tools. The chatbot was developed using a version of GPT-4, OpenAI's Chat Assistant API, and retrieval-augmented generation (RAG) via OpenAI's Embeddings API. It incorporated a ground-truth external data source derived from the course syllabus to enhance context relevance and was prompted to guide students rather than giving away the answer. In class, the instructor did not provide explicit limitations on how to use the chatbot.

To understand student-level constructs that correlate with chatbot queries, we gave students a survey that assessed their *goal orientation* (Adesope et al., 2015), *task value* (Pintrich et al., 1993), and views on *LLM permissibility* in future academic and professional settings (e.g., "In your opinion, how permissible will it be to use LLM technologies in your future work/career settings?"). The survey also included eleven scenario-based questions about chatbot usage and students were asked to rate the appropriateness of these usage behaviors for learning. For example, in a scenario labeled as "*Requesting solution for pset step*," the survey asked, "A student asks the chatbot for help with a specific programming task. They prompt the chatbot, 'How do I...' and paste in step 3 from their problem set. How appropriately is this student using the chatbot to support their learning?" These scenario-based questions were designed to assess student *perceptions* of appropriate chatbot queries to support learning.

In total, students prompted the chatbot 2,731 times throughout the semester. We collected survey data from $N = 107$ students in the course who also consented to their chat data being used for research. Of these students, $n = 57$ used the chatbot at least once for help on their problem set, resulting in a total of 855 queries to the chatbot. The first author and a second coder qualitatively coded these chats according to the scenarios presented in the survey (e.g., *Requesting solution for pset step*), achieving a final inter-rater reliability of $k=0.78$.

At the aggregate level, students perceived "*Requesting a conceptual explanation*" as the most appropriate type of query, and "*Requesting solution for pset step*" as the least appropriate. In our preliminary analysis, we used Pearson's correlation to correlate students' goal orientation, task value, and LLM permissibility with the number of times students asked these two categories of questions to the chatbot. Our findings showed that students with a higher *mastery-approach* orientation ($r(57) = 0.35, p < .01$) and those who placed a higher *task value* on the course material ($r(57) = 0.27, p < .05$) requested more conceptual explanations from the chatbot (e.g., "What is recursion and why would I use it?"), while students with a higher *work-avoidant* orientation asked fewer conceptual questions ($r(57) = -0.28, p < .05$). This suggests that students' motivational constructs may influence their question-asking behavior. Further, we found that students who thought LLM use would be widely permitted in their future school or work settings rated the "*Requesting solution for pset step*" scenario as more appropriate ($r(107) = 0.27, p < .01$). As one student reasoned in the free-response portion of the survey, "*banning the use of AI is kind of silly...when my work asks that I code something in the future, odds are, I will use AI to do so.*" Students who rated this scenario as more appropriate also requested more direct code solutions from the chatbot ($r(57) = 0.59, p < .001$). Although this "over-reliant" behavior raises concerns, these findings suggest that clarifying norms around appropriate chatbot use and balancing course learning goals with students' beliefs about the use of LLMs in future contexts may support more self-regulated chatbot usage in class.

Contribution and future directions

Our preliminary findings shed light on how students' motivational constructs and perceptions of appropriate chatbot use correlate with actual chatbot help-seeking behaviors in an introductory CS course. The findings highlight the importance of instructional framing: (1) explicitly discussing appropriate chatbot use and (2) aligning course learning goals with students' perceptions of LLM use in future contexts. Future work could explore how, in addition to comprehensive guardrails, chatbots can help co-regulate student behavior and promote more productive, learning-oriented interactions.

Contribution 3#: Learning with and about AI through LLM-supported interdisciplinary collaborative design

Carolyn P. Rosé, Melisa Orta-Martinez, Kylie Peppler

Introduction

We investigate new ideas in human-AI collaboration through a Learning by Design approach (LDB; Hmelo et al., 2014; Kolodner et al., 1998), motivated by our own curriculum development and teaching of Applied Machine Learning for over two decades as well as over 15 years of research on creating high-quality STEAM materials

and published curricula for makerspaces, K-12 settings, and higher education (Peppler et al., 2014; Peppler et al., 2023). LBD has been acknowledged as having an important connection to AI literacy (Touretzky et al., 2023).

Ongoing work

We present an investigation into curriculum development to offer students a 3-week transformative encounter with the limitations of current LLMs based on insights with respect to abstraction and reasoning, limitations that are very clear to us in our own technical research on advancing LLM capabilities in automated code translation (Xie et al., 2023) and financial question answering (Nourbakhsh et al., 2022) as well as work on an established benchmark task called the ARC Challenge dataset, designed to highlight the limitations of large language models (LLMs) with respect to abstraction and decomposition. One major challenge with introducing concepts regarding the connection of representations and reasoning in machine learning is the advanced level of math and abstract reasoning required to understand those concepts deeply. We address this through making the abstract concepts concrete through hands-on design activities in the textile space, where the design of a physical artifact is investigated through mathematical representations at multiple levels of abstraction and tied to the key idea of matrix factorization, considered to be the foundation for all of machine learning. Activities and assessments engage and probe student reasoning in the mapping between representations. Thus, the activity offers students the opportunity to demonstrate their ability at abstraction and decomposition as they collaboratively construct solutions together. And although LLMs are limited in their ability to perform abstraction and decomposition on their own, LLMs have been successful in their instantiation as support agents in human-human collaborative learning in our own past work where the humans are performing the problem solving, and an LLM agent offers feedback and reflection prompts related to design principles and how they might be applied in context (Naik et al., 2024), with demonstrated positive impact on student learning.

As learners explore a design space afforded by a 16-shaft loom, they are meant to realize that in decomposing a matrix into its component matrices, it is possible to identify that alternative surface designs may incorporate a common transformation or the same underlying profile block element. Examples like this provide the opportunity to observe limitations in neural models, and particularly LLMs, with respect to hierarchical decomposition. In particular, whenever multiple design elements – such as a block substitution followed by another block substitution, such that the design is created through a hierarchy of transformations operating at different levels, only the surface level of structure is recognized by the LLM. Even for humans looking at a textile design, the last transformation is the most obvious since that is what determines the texture that hits the eye first. We find the same behavior in our work on applying LLMs to a range of benchmark tasks. As students engage in design behavior, as they apply successions of weaving techniques, they are enacting a hierarchical algorithmic process, as though they are generating behavior that has a hierarchical decomposition into a representation. While these design practices can be composed into a seemingly limitless landscape of design potential, the physicality of a loom places constraints on the representational capacity of those matrices in a very similar way to what creates capacity limitations on machine learned models. Compression and composition are especially key to achieving complex designs in the face of capacity limitations, just as in neural models (e.g., Vincent et al., 2008). An important insight related to the connection between limitations in AI capabilities and representations is to note how they arise due to the extreme need for compression (resulting in information loss) and the challenges related to decomposing highly compositional representations (resulting in overfitting to trivial patterns and thus missing important nuances).

Contributions and future directions

In response to the need for more insights for teaching the connection between machine learned representations and reasoning-like behavior in AI agents, we introduce the computer-supported collaborative learning activity design here, in preparation for a Spring 2025 deployment in one university level class as a foundation for proposing future studies in this under-studies and critical space (findings to be reported at the symposium). In particular, we have created a design activity using a weave structure called pique that allows weaving a quilted object all in one piece. The underlying multi-layer modeling embodying a specific abstraction and decomposition over the design space, which facilitates turning a design idea into a woven potholder built on top of a foundational 16-shaft pique threading, is embodied within a Python program that students can interact with in a Jupyter Lab environment. Students will interact with an LLM agent during their collaborative design and will have the opportunity to ask questions and probe into “machine perception” as the Jupyter environment has been set up as a computer-supported collaborative learning environment with both well-established Bazaar collaborative learning support agents (Adamson et al., 2014) and newer LLM support agents (Naik et al., 2024). Through jointly analyzing chatlogs and properties of student constructed representations at different levels of abstraction, we will investigate student thinking both in their individual and collaborative work.

Contribution 4#: Augmenting socially shared regulation of learning with a collaborative conversation agent

Joni Lämsä, Adelson de Araujo, Justin Edwards, Marta Sobocinski, Anni-Sofia Roberts, Ridwan Whitehead, Pantelis M. Papadopoulos, Sanna Järvelä

Introduction

Success in computer-supported collaborative learning (CSCL) tasks requires socially shared regulation of learning (SSRL; Hadwin et al., 2017). In SSRL, learners work together to understand the task, set common goals, plan and enact strategies to reach those goals, and adapt their understanding and strategies if their progress is not on track. These SSRL phases—task understanding, setting goals and planning, enactment, and adaptation—are particularly important in collaborative inquiry tasks when students need to discover further information themselves (de Jong et al., 2024). However, collaborative inquiry tasks may pose various cognitive and metacognitive obstacles that may be triggers for SSRL, such as having difficulties in making sense of the task (Järvelä & Hadwin, 2024). To develop real-time support for SSRL, recent advancements in artificial intelligence (AI) have enabled novel ways to detect obstacles in CSCL (Suraworachet et al., 2024). Since SSRL necessitates jointly negotiated goals and (the enactment of) joint task plans that may be adapted in response to these obstacle events, collaborative conversation agents (Edwards et al., 2025) could augment SSRL by helping students to share their thoughts, orient to and listen carefully to one other, deepen their reasoning, and engage with others' reasoning that are known as four goals for productive dialogue (Michaels & O'Connor, 2015). Thus, we aim to investigate how group engagement in socially shared regulation of learning (SSRL) and dialogue productivity evolve in response to prompts from a collaborative conversational agent during a collaborative inquiry task.

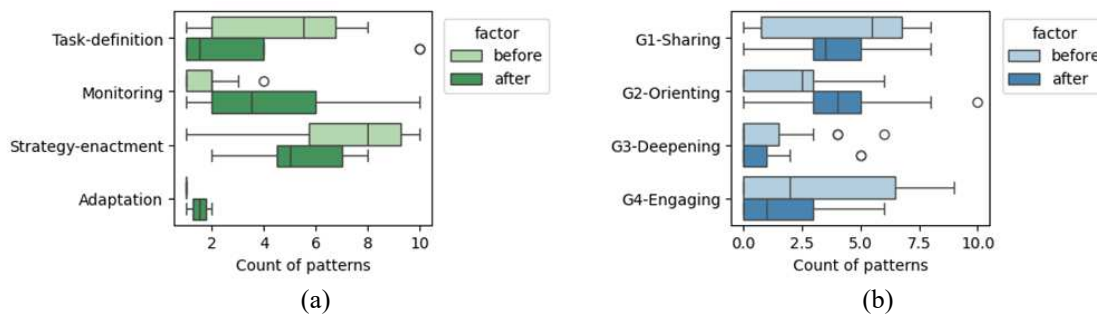
Ongoing work

$N = 52$ preservice teachers were divided into 14 groups of 3–4 members and worked face-to-face. This study focuses on a 20-minute collaborative inquiry physics task on energy. A metacognitive AI (MAI) agent (Edwards et al., 2025) listened to the groups' discussions and invited SSRL via a loudspeaker (e.g., "I noticed you might be confused. Can you explain what you're trying to understand right now?") if it noticed one of the following obstacle events: 1. Students used the keywords "I/we don't/can't ..." three times consecutively, indicating that students were monitoring their lack of progress; 2. Students expressed confusion three times consecutively, indicating conceptual challenges; 3. Students used the first or second person singular ("I"/"you") three times consecutively, indicating a lack of shared perspective. The 14 groups received a total of 18 MAI prompts during the task. To measure engagement in the SSRL phases, we video recorded, transcribed, and coded the groups' discussions into task understanding, strategy enactment, adaptation, and monitoring of learning (Cohen's $\kappa = 0.57$ – 0.71 , Cohen, 1960). To measure the four goals for productive dialogue (Goal (G) 1: sharing thoughts, G2: orienting to and listening carefully to one other, G3: deepening reasoning, and G4: engaging with others' reasoning), we used the *focus* and *intent* of utterances identified by a machine learning-based system for automated coding of student discussions (de Araujo et al., 2024). By applying sequential pattern mining to the focus and intent of utterances, we tracked key behaviors and determined outcomes based on the frequency of patterns that indicated those behaviors. The pattern conditions specifically employed the following proxies to operationalize the four goals: G1 identifies sequences of task-related informative and argumentative statements, indicating thought sharing; G2 recognizes responses to peers' task-related questions; G3 emphasizes expanding one's own task contributions; and G4 focuses on collaborative argument discussion. Thresholds were set using each dialogue variable's high, medium, and low limits from the triggering mechanism. We compared the presence of the SSRL phases and four goals for productive dialogue of 20 utterances before and after MAI's prompts.

Paired t -tests showed that monitoring occurred more frequently ($t(17) = 2.96$, one-sided $p = .004$, Figure 1a) after MAI's prompts ($M = 3.6$, $SD = 2.7$) than before ($M = 1.4$, $SD = 1.2$). Consequently, strategy enactment occurred less frequently ($t(17) = -1.89$, one-sided $p = .038$) after MAI's prompts ($M = 4.7$, $SD = 2.7$) than before ($M = 6.4$, $SD = 3.6$). Orienting and listening carefully to one another were more frequent ($t(17) = 2.32$, one-sided $p = .017$, Figure 1b) after MAI's prompts ($M = 4.1$, $SD = 2.6$) than before ($M = 2.2$, $SD = 2.1$).

Figure 1

Mean Frequency of the Utterances (a) Coded into the Different SSRL Phases and (b) Having the Different Goals (G) for Productive Dialogue Before and After MAI's Prompts



Contribution and future direction

Advancements in AI have shown promise in supporting SRL during individual inquiry tasks (Liu et al., 2024), but there is a paucity of research focusing on SSRL within collaborative inquiry tasks (Sharma et al., 2024). We addressed this gap by showing how a collaborative conversation agent MAI's prompts rarely led groups to adapt their task understanding or enacted strategies, but those triggered monitoring of learning. As orienting to and listening carefully to one another was also more frequent after the MAI's prompts, our findings suggest that interactions among students and MAI can facilitate students' joint negotiation and evaluation of the ongoing learning process and progress. Considering the short duration of the task in this study, future research should focus on longitudinal studies that capture the trajectories of human-AI collaboration patterns, and how these patterns foster agentic learners who engage in SSRL and productive dialogue in collaborative inquiry tasks.

Significance of this symposium

In concluding, we would like to emphasize that this symposium aims to advance the conversation on hybrid intelligence in teaching and learning. The presented papers offer early empirical insights and highlight the need for further research into the asymmetries within human-AI relationships. They demonstrate that the strengths and limitations of both human learners and AI vary depending on specific tasks and contexts, as learning is inherently situated. Moreover, the evolving nature of human-AI interactions underscores the need for continued conceptual development—whether framed as co-learning, co-evolution, or complementarity.

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